

WASP-12b

R-0562

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_170204 · created 2026-07-23T18:16:30+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457788.69376 +/- 0.00036 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.0 +/- 0.028
Transit depth (Rp/Rs)^2	4.1e-06 +/- 0.0011 [%]
Orbital Inclination (inc)	79.06 +/- 2.83
Airmass coefficient 1 (a1)	1.054 +/- 0.039
Airmass coefficient 2 (a2)	-0.049 +/- 0.085
Scatter in the residuals of the lightcurve fit is	3.68 %
Best Comparison Star	#2 - [154, 388]
Optimal Aperture	14.95
Optimal Annulus	23.35
Transit Duration (day)	0.097 +/- 0.015

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.0 ± 0.028 vs 0.1178 (deviation 4.01σ)
Duration fit vs published	0.097 d vs 0.125 d (deviation 1.78σ)
Mid-time O–C	0.3 min
Depth SNR	0.0
Residual scatter	3.68 %

Light curve

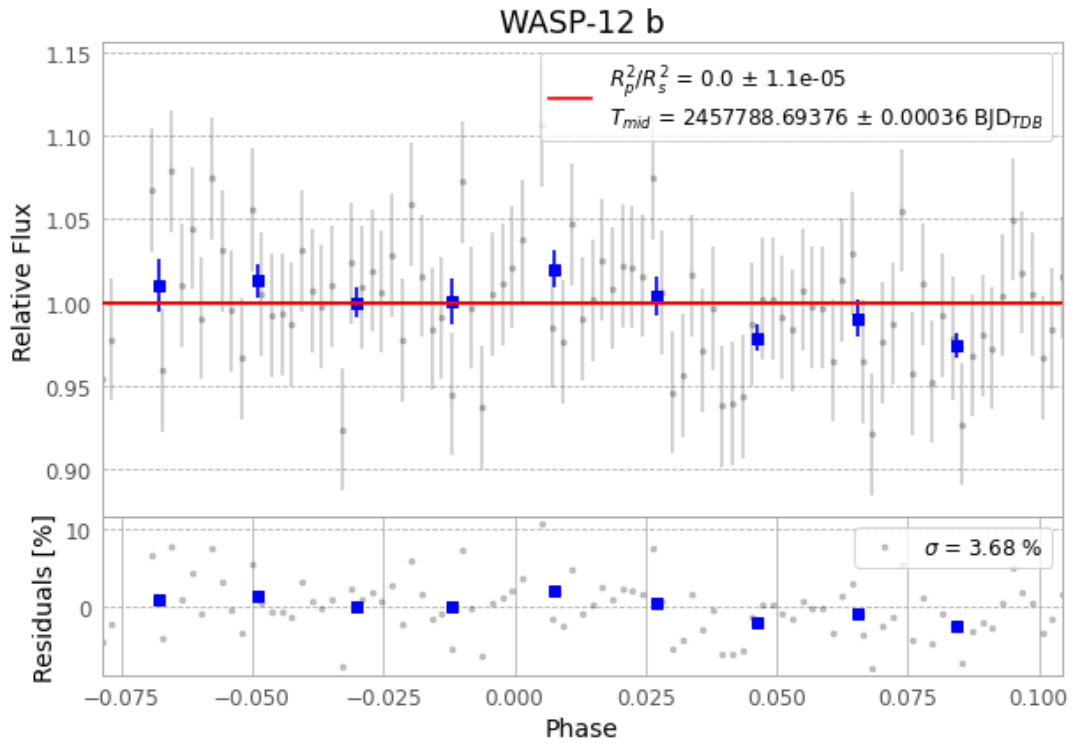


Figure 1 — FinalLightCurve_WASP-12 b_2017-02-03.png (SHA-256 ed722316af9e3a0b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [348, 344], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 3f9d11729dfa8bde...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

97 FITS frames (64 MB), WASP-12170204022712.FITS → WASP-12170204071512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-170204.log (84da9322d528...), FinalLightCurve_WASP-12 b_2017-02-03.png (ed722316af9e...), FinalLightCurve_WASP-12 b_2017-02-03.csv (afaed345de88...)

Compute resources

Wall time 340.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-23 18:16Z.